CHAPTER 4

MACROSCOPIC AND MICROSCOPIC STATES

The treatment given in the preceding chapter was limited to the problems connected with the properties of ideal gases. We now proceed to describe a more general method for dealing with the problems in statistical physics. The method can be applied to a system much more complicated than a gas system, and of macroscopic dimensions, whose constituent particles may interact with one another.

A system is generally characterized by the physical and chemical properties of the substance in the space occupied by it, as well as by the properties of boundaries. In general, we are concerned with systems consisting of a large number of particles, such as for example, a gas comprising a large number of molecules contained in a vessel; or a crystal consisting of a large number of atoms in a lattice etc. The properties of a system depend upon the state in which it exists at the instant at which measurements are made. Our first concern, therefore will be how to specify the state of a system at any time.

4.1 MACROSCOPIC STATES

Logically since all system obey quantum mechanics we should start with quantum statistical physics and then arrive at classical statistical physics as a particular case. But it is not always necessary to use quantum mechanics in describing the behaviour of a thermodynamic system. Newtonian mechanics is in excellent approximation and, hence it, convenient to commence the study of statistical mechanics consideration.

To explain the concept of the state of a system, we will once more refer to a monatomic ideal gas, which is relatively a simple system. By an ideal gas we mean that its constituent particles have a finite mass, undergo elastic collisions with one another and with the boundary walls and have no other interaction of any kind. We further assume that the vessel containing the gas is isolated. That is, it does not exchange energy with the material bodies outside and, hence, all variations in the gas are due to internal reasons. The time spent in collision is very much shorter than the time lapse between collisions. It must be mentioned here that real gases are not ideal in the sense we have described above. They approximate to the ideal gases when the pressure is very low. Under this condition the particle density is so low that the particles are relatively distant from each other. If a gas is left to itself for a sufficiently long time, the pressure and the temperature of the gas, whatever their initial distribution, stabilize over the entire volume and we say that the gas has attained a steady state. In this state the pressure (P), temperature (T) and volume (V) remain constant in time. This state of the gas, characterized by the parameters, P, T and V, is called its *macroscopic state*. The macroscopic steady state in this particular case is also called its equilibrium state. Note that a steady state is not always an equilibrium state. For a system that is not isolated, it is possible to establish a steady state by maintaining the different parts of the walls of the vessel at different temperatures with the help of external sources. We will however, restrict ourselves in the first few chapters to the study of equilibrium state, as these provide a great simplification. Besides most systems we encounter are in thermodynamics equilibrium or else they tend to reach very rapidly to a state of equilibrium.

4.2 MICROSCOPIC STATES

Since a gas consists of a large number of particles, the state of a gas can also be characterized in terms of the states of the constituent particles. The states of the particles can be specified by ascertaining their positions (coordinates q_i) and velocities or momenta p_i. The state of the gas thus described in terms of the properties of its constituent particles is called its *macroscopic state*.

We have seen above that the quantities pressure, temperature and volume which describe the macroscopic state remain constant in the steady state of the gas. The constituent particles of the gas, however, are continuously in motion and undergo collisions even in a steady state and hence microscopic state keep on changing continuously. Let us be more clear on this point. The microscopic state of a gas at any instant is defined by the instantaneous positions and momenta (in magnitude and direction) of the various molecules. If any change is made in the molecular distribution, a new microscopic state results. Many of the different microscópic states, however, will be indistinguishable from the standpoint of macroscopic observation. For instance, if the positions and momenta of two or more molecules are interchanged, a new microscopic state will result; yet from the macroscopic point of view, no change will be detected in the condition of the gas. Each one of the microscopic states is said to pertain to the same macroscopic state. We are thus led to associate with each microscopic state a large number of microscopic states, although they widely differ in respect of their dynamical states.

Consider, for example, an ideal gas system in which particles can exist in any one of the states with energy equal to 0, 1, 2, 3 energy units and no other. Suppose the system of three distinguishable particles a, b and c and the total energy of the system is three energy units. In how many ways three energy units can be distributed among the three distinguishable particles, consistent with the condition that the total energy of the system is three units? In other words, how many microscopic states correspond to the macroscopic state specified by three energy units?

Figure 4.1 shows the possible configurations of the system and the number of microscopic states.

Configuration	Number of states	
	Distinguishable particles	Indistinguishable particles
3: *a* | *b* | *c* 2: | | 1: | | 0: *bc* | *ac* | 2*b*	3	1
3: *a* | *a* | *b* | *b* | *c* | *c* 2: *b* | *c* | *c* | *a* | *a* | *b* 1: 0: *c* | *b* | *a* | *c* | *b* | *b*	6	1
3: 2: 1: 0: *abc*	1	1
Total number of states	10	3

Fig. 4.1

The total number of accessible microscopic states corresponding to the single macroscopic states in ten of the particles are distinguishable and three if they are indistinguishable.

One of the aims of statistical mechanics is to correlate the properties of matter in bulk with the properties of its constituent particles i.e. to establish a relation between a macroscopic state and the corresponding microscopic states.

4.3 PHASE SPACE

We shall now show how the state of a system is specified. We assume that the physical systems obey the most general laws of dynamics as set down by Hamilton. The state of a system is determined when we know the details of its motion and, hence, is fully specified at a particular time by a certain number of generalized coordinates q_i and an equal number of generalized momenta p_i. These coordinates and momenta are governed by the canonical equation of Hamilton:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}; \ \dot{p}_i = -\frac{\partial H}{\partial q_i} \quad (i = 1, 2, ...) \qquad ...(4.1)$$

where q_i are the generalized coordinates

p_i are the generalized momenta

and H is Hamiltonian of the system

A most elementary physical realization of such a system is furnished by a single particle constrained to move in a straight line. The position of the particle at any time can be described by giving its distance from a fixed point on the line—origin of coordinates. This single numerical fact merely gives its 'mechanical configuration' at the time in question. For the complete specification of its state, its momentum at this instant must also be given. Since in elementary physics mass is assumed to be constant momentum is proportional to the velocity and, hence, it may seem that it matters little which of the two is used. However, it must be emphasized that the generalized momenta used in Hamiltonian equations are very different from the mass—velocity products of elementary mechanics.

The state of a particle at any time is given by the pair of values (q, p) = (position, momentum) determined by the differential equations of dynamics [Eq. (4.1)]. As the point moves on a straight line, the values of number pair (q, p) will change. Quasi-geometrical terminology makes for convenient and lucid expression and considerably facilitates the study of the changes in the state as time goes on. Any number pair can be plotted as the rectangular coordinates of a point in a Cartesian graph (Fig. 4.2).

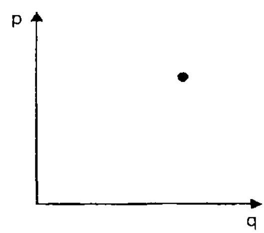

Fig. 4.2

Every such point which corresponds to a certain value of coordinate q_i, and momentum p_i of the system, describes both the position and momentum of the system at a particular instant, and hence represents a certain state of the system.

There will be different points for different states. As the particles moves on the straight line, its momentum and coordinate vary in accordance with Hamilton's equation [Eq. (4.1)]. Its evolution through the successive states will be represented by a trajectory in the Cartesian plane (Fig. 4.3).

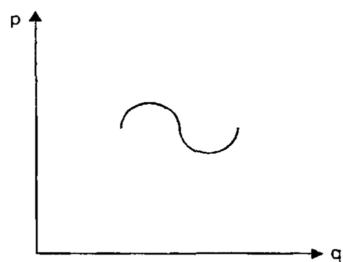

Fig. 4.3

The state of the system when represented in this way is referred to as its *phase*, the representative point as the *phase point*, the trajectory as the *phase path* and the Cartesian plane as the *phase space*.

It is not necessary to start with Cartesin coordinates. Polar cylindrical or any other suitably defined set of coordinates would do as well. Nevertheless, we shall find that the phase space with rectangular axes corresponding to each coordinate and momentum is very convenient, as it has simple properties.

Example 4.1 Determine the phase trajectory of a bullet of unit mass fired straight upwards with an initial speed of 392m/s. Acceleration due to gravity is approximately 9.8m/s^2 or,

$$\frac{dv}{dt} = -9.8$$

Therefore $\quad v = -9.8\ t + C_1$ (a constant)

From boundary conditions viz.

$$v = 392 \text{ when } t = 0,\ C_1 = 392$$

Therefore $\quad v = 9.8\ t + 392 \qquad$ ...(4.2)

$$p = mv = -9.8\ t + 392 \qquad \text{(because } m = 1\text{)} \qquad \text{...(4.3)}$$

Now $\quad v = \dfrac{dq}{dt} = -9.8\ t + 392$

On integrating $q = -4.9\ t^2 + 392\ t + C_2$ (a constant)

Again, using intial conditions $q = 0$ when $t = 0$, $C_2 = 0$

Therefore, $\quad q = -4.9t^2 + 392\ t \qquad$ ...(4.4)

Equations (4.3) and (4.4) give the pair values of the phase point at different times. Thus

(*a*) when	$t = 0$,	$q = 0$,	$p = 392$
(*b*) when	$t = 10$,	$q = 3430$,	$p = 294$
(*c*) when	$t = 30$,	$q = 7350$,	$p = 98$
(*d*) when	$t = 40$,	$q = 7840$,	$p = 0$
(*e*) when	$t = 60$,	$q = 5880$,	$p = -196$
(*f*) when	$t = 80$,	$q = 0$,	$p \doteq -392$

If we plot these points as shown in Fig. 4.4, we get the phase space trajectory for the phase point of the bullet. The trajectory is obviously a parabola, the equation of which can be obtained by eliminating t between Eqs. (4.3) and (4.4) i.e.

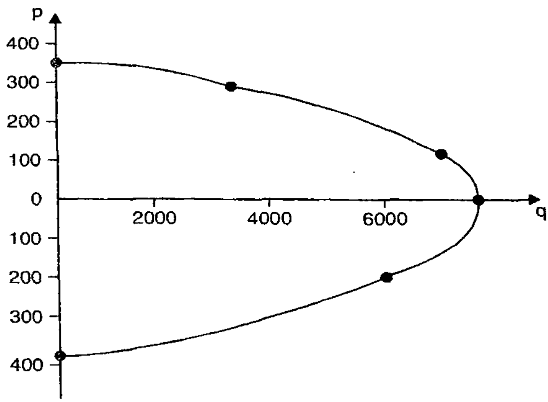

Fig. 4.4

$$q = -\frac{p^2}{196} + \frac{(392)^2}{19.6}$$

or
$$p^2 + 19.6\,q - 153664 = 0 \qquad ...(4.5)$$

Example 4.2 A particle of unit mass is executing simple harmonic vibrations. Determine its trajectory in phase space.

The total energy of the particle E which is constant of motion is given by

$$E = \frac{p^2}{2} + \frac{1}{2}\,kq^2 \qquad ...(4.6)$$

where the first term represents its kinetic energy and the second its potential energy.

The above equation can be written as

$$\frac{p^2}{2E} + \frac{q^2}{2E/k} = 1 \qquad ...(4.7)$$

The changes in q, p will always be subject to this condition. Equation (4.7) is the equation of the curve representing the successive states (phase) of the vibrating system. The equation represents an ellipse (Fig. 4.5) with semi-axes equal to $\sqrt{2E}$ and $\sqrt{2E/k}$. Any point on the ellipse represents the phase (q, p) at some time, and the ellipse is described over and over again, since the particle is vibrating periodically in a straight line, representing the same positions and momenta.

For a given pair (q, p) the Eqs. (4.1) determine uniquely the rate of change of every coordinate in phase space. It follows at once that through every point in phase space, there passes but one trajectory. Hence, the representative point can never cross its previous path.

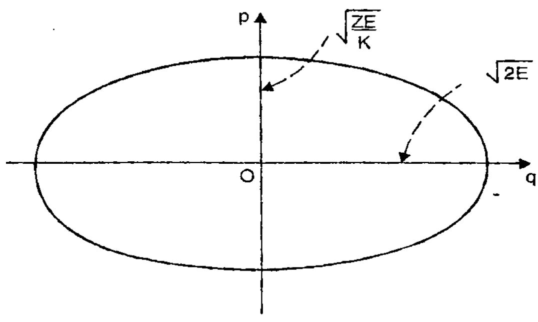

Fig. 4.5

Thus, the locus of a representative point in phase space is either a simple closed curve, or a curve that never intersect itself.

In the two examples that we have considered, it may be noted, that the actual motion of the system takes place in a straight line. The parabola in the first example and ellipse in the second, is only the progress chart, not the actual path of the moving particle.

4.4 THE μ-SPACE

A mechanical system slightly more advanced than a particle moving in a straight line, is a monatomic molecule moving in space. In a three-dimensional configuration space, the position of molecule is determined by the coordinates x, y, z. These coordinates, however, do not say anything about its motion. If instead, we imagine a three-dimensional space in which the coordinates are the momentum projections p_x, p_y, p_z a momentum space, a point in it determines the state of motion of the molecule, but nothing about its position. A very meaningful way to determine the complete state of a molecule would be to set up a six-dimensional space with coordinates x, y, z, p_x, p_y, p_z in which every point represents a state of the molecule. Such a six dimensional space for a single particle is called its *μ-space*. Note that phase space is purely a mathematical concept.

Suppose that the position or momentum of one of the molecules is changed slightly. The point of this molecule will undergo a displacement in the phase space, and the microscopic state of the gas will be modified. However, if the changes in the position and momentum of the molecule so slight that even the most accurate experiment would be unable to reveal them, we should assume that no change in the microscopic state has occurred.

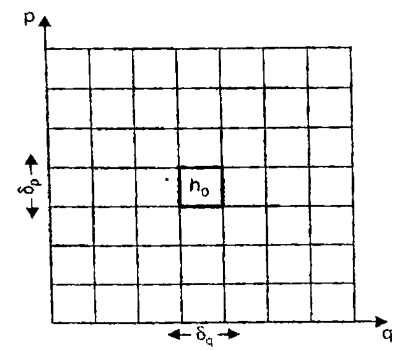

Fig. 4.6

Graphical representation of the foregoing is easily obtained. Consider a two-dimensional case. We subdivide the range of variables q and p into arbitrarily small discrete intervals. Let δq be the size of the intervals on the q-axis and δp, the corresponding size on the p-axis. Phase space is thus divided into small cells of equal area $= \delta q \delta p = h_0$ (say) (Fig. 4.6). The state of the system can then be specified by stating that the representative point lies in a particular cell of phase space. In classical description h_0 can be chosen arbitrarily small and the specification of state becomes more and more precise, as we choose smaller and smaller magnitudes for h_0. It may be noted that h_0 has the dimension of energy × time. The lengths δq and δp are assumed to be so small that if a phase-point within a cell is slightly displaced to any point within the cell, the change in the position and the momentum will to be too minute to be disclosed even by accurate measurements. If the particles is in the cell, its state is represented by the pair of coordinates (q_i, p_i). If there is a change in the state of the particle it will move from this cell to another appropriate cell.

4.5 THE G-SPACE

For a system with n atoms a set of 6 n members $(q_1, q_2, \ldots q_{3n}, p_1, p_2, \ldots p_{3n})$ is required to represent a phase point. Such a $6n$-dimensional space is called Γ-*space*. *This* is evidently built up of n spaces of μ-type. It is also sometimes more convenient to regard Γ-space as constructed out of the *configuration space* that corresponds to a set of $3n$ momenta conjugate to the coordinates.

In $6n$-dimensional case, we imagine, as in the two-dimensional case, the phase space to be partitioned into tiny juxtaposed 6-dimensional cells. These cells are all equal, the edges of the cells being of equal magnitude. We define an element of volume in Euclidean space $dv = dx\,dy\,dz$ so that a system with coordinates within the range between x and $x + dx$, y and $y + dy$, z and $z + dz$ lies within it. Analogously, a volume element in $6n$-dimensional phase space is expressed as

$$\begin{aligned} d\Gamma &= dq_1\, dq_2 \ldots dq_{3n}\, dp_1\, dp_2 \ldots dp_{3n} \\ &= h_0^{3n} \end{aligned} \qquad \ldots(4.8)$$

and the state of the system can be specified by starting in which particular cell the coordinates $q_1, q_2, \ldots q_{3n}, p_1, p_2 \ldots p_{3n}$ of the system are found. The succession of states will be represented by trajectory in the $6n$-dimensional space. Such a space is difficult to visualise. It is purely a mathematical concept. In general, therefore, one deals with abstract paths in an abstract space.

It will be observed that the f-dimensional hyper volume in the phase space has the dimensions of (energy × time)^f. These dimensions hold regardless of the coordinates used. It can also be shown that on change of variable any given hyper volume goes over to a hyper volume of equal size in the new phase space (Prob. 4.3).

In Boltzmann's days there was no theoretical reason to ascribe any specific value h to the volume element in the phase space. But with the discovery of the Planck constant and the advent of quantum theory, more light was thrown on the situation. We see that the assumption $\delta q \delta p = h$ is now consistent with the Heisenberg uncertainty relation. According to this relation, if the position of a particle is uncertain within a volume dx dy, dz, the momentum components of the particle are uncertain within a volume dp_x, dp_y, dp_z of momentum space. Consequently, all particles situated in the special volume $dx\,dy\,dz$ are indistinguishable. And between these two volumes we have the relation

$$dx\,dy\,dz\,dp_x\,dp_y\,dp_z = h^3 \qquad ...(4.9)$$

4.6 POSTULATE OF EQUAL A PRIORI PROBABILITIES

As stated in Chapter 2, it is convenient to analyze various problems in statistical physics, if we follow the method of *ensembles*. A vessel containing an ideal gas is a statistical system. An aggregate of such identical systems such as above is called a *statistical ensemble*, following the original terminology of Gibbs, one of the founders of statistical mechanics. Since the systems of the ensemble are identical their macroscopic states are the same, but microscopic states may be different.

Consider a system composed of N identical particles in a volume V, n_1 of which are with energy ε_1, n_2 with energy ε_2 etc. If the particles are non-interacting the total energy is given by

$$E = \sum_i n_i \varepsilon_i \text{ and } N = \sum_i n_i \qquad ...(4.10)$$

The specification of the actual values of the parameters E, V, N then defines a particular 'macrostate' of the given system. Since the total energy E consists of a simple sum of energies ε_i, there will obviously by a large number of different ways in which the individual ε_i can be chosen, so as to make the total energy E. In other words, there will be a large number of different ways in which the total energy equal to E of the system can be distributed among the N particles constituting it. Each of these different ways specifies a particular microstate of the given system. In any case, to a given macrostate of the system there does, in general, correspond a large number of microstates and it seems natural that at any time t, the system is equally likely to be in any one of these microstates. That is, all microstates are equally probable. This is generally referred to as the *'principal of equal a priori probability'*. This principal can be considered as the basis of the entire theory of systems in equilibrium. If all the microscopic states have the same probability, the probability of a given macroscopic state will be proportional to the number of different states which it contains.

4.7 ERGODIC HYPOTHESIS

In statistical mechanics we have often to deal with the average or the mean of a quantity. The average of a physical quantity can be determined in two ways: (*i*) one could consider an ensemble of a large number of identical systems and average, the physical quantity over all these systems at one instant of time to determine its *ensemble average*; or (*ii*) a system could be followed over a very long period of time, during which the system takes different values. The average of the physical quantity over the long periodic gives the *time averaged* value of the quantity.